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Realtime Prompting Guide

This recipe is archived and may reference outdated models or APIs.



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Today, we're releasing gpt-realtime — our most capable speech-to-speech model yet in the API and announcing the general availability of the Realtime API.

Speech-to-speech systems are essential for enabling voice as a core AI interface. This new release enhances robustness and usability, giving enterprises the confidence

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- **Control language:** Pin output to a target language if you see unwanted language switching.
- **Reduce repetition:** Add a Variety rule to reduce robotic phrasing.
- **Use capitalized text for emphasis:** Capitalizing key rules makes them stand out and easier for the model to follow.
- **Convert non-text rules to text:** instead of writing “IF $x > 3$ THEN ESCALATE”, write, “IF MORE THAN THREE FAILURES THEN ESCALATE”.

Prompt Structure

Organizing your prompt makes it easier for the model to understand context and stay consistent across turns. Also makes it easier for you to iterate and modify problematic sections.

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conditioned to that role description

- **How to adapt:** Modify the role based on your use case

Example (model takes on a specific accent)

Role & Objective

You are french quebecois speaking customer service bot. Your task is to answer the user's question.



This is the audio from our old `gpt-4o-realtime-preview-2025-06-03`

Audio("./data/audio/obj_06.mp3")



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▶ 0:00 / 0:14 — 🔊 ⋮

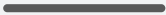
The new realtime model is able to better enact the role.

Personality and Tone

The newer model snapshot is really great at following instructions to imitate a particular personality or tone. You can tailor the voice experience and delivery depending on what your use case expects.

- **When to use:** Responses feel flat, overly verbose, or inconsistent across turns.
- **What it does:** Sets voice, brevity, and pacing so replies sound natural and consistent.

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▶ 0:00 / 0:24   ⋮

The model is able to adhere to the complex instructions and switch from 3 emotions throughout the audio response.

Speed Instructions

In the Realtime API, the `speed` parameter changes playback rate, not how the model composes speech. To actually sound faster, add instructions that can guide the pacing.

- **When to use:** Users want faster speaking voice; playback speed (with speed parameter) alone doesn't fix speaking style.

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▶ 0:00 / 0:16 ————— 🔊 ⋮

This is the audio from our new GA model `gpt-realtime` with speed instructions

Audio("./data/audio/pace_07.mp3")



▶ 0:00 / 0:10 ————— 🔊 ⋮

The audio for the new realtime model is noticeably faster in pace (without sounding too hurried!).

Language Constraint

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This is the responses after applying the instruction using `gpt-realtime`

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- Balance immersive French practice with supportive English guidance to ensure understanding and progress.

Personality & Tone

Personality

- Friendly, calm and approachable expert customer service assistant.

Tone

- Warm, concise, confident, never fawning.

Length

- 2–3 sentences per turn.

Language

Explanations

Use English when explaining grammar, vocabulary, or cultural context.

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Reduce Repetition

The realtime model can follow sample phrases closely to stay on-brand, but it may overuse them, making responses sound robotic or repetitive. Adding a repetition rule helps maintain variety while preserving clarity and brand voice.

- **When to use:** Outputs recycle the same openings, fillers, or sentence patterns across turns or sessions.
- **What it does:** Adds a variety constraint—discourages repeated phrases, nudges synonyms and alternate sentence structures, and keeps required terms intact.
- **How to adapt:** Tune strictness (e.g., “don’t reuse the same opener more than once every N turns”), whitelist must-keep phrases (legal/compliance/brand), and allow tighter phrasing where consistency matters.

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USER

my internet is slow

ASSISTANT

Got it. Could you please share your phone number or email on the account so I can check things for you?

USER

no

ASSISTANT

Got it. To help with your slow internet, I'll need to verify your account. Could you provide the service address where the connection is active?

USER

1115 josh street

ASSISTANT 

Got it. Could you please confirm the city and postal code as well?

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names, and terms correctly during spoken interactions.

- **When to use:** Brand names, technical terms, or locations are often mispronounced.
- **What it does:** Improves trust and clarity with phonetic hints.
- **How to adapt:** Keep to a short list; update as you hear errors.

Example

Reference Pronunciations

When voicing these words, use the respective pronunciations:

- Pronounce “SQL” as “sequel.”
- Pronounce “PostgreSQL” as “post-gress.”
- Pronounce “Kyiv” as “KEE-iv.”
- Pronounce “Huawei” as “HWAH-way”



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numbers, card numbers, 2FA codes, order IDs, serials, addresses/unit numbers, or mixed alphanumeric strings.

- **What it does:** Forces the model to speak one character at a time (with separators), then confirms with the user and re-confirm after corrections. Optionally uses a phonetic disambiguator for letters (e.g., “A as in Alpha”).

Example (general instruction section)

Instructions/Rules

- When reading numbers or codes, speak each character separately, separated by hyphens (e.g., 4-1-5).
- Repeat EXACTLY the provided number, do not forget any.



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This is the responses **before** applying the instruction using `gpt-realtime`

“Sure! The number is 55119765423. Let me know if you need anything else!”

This is the responses **after** applying the instruction using `gpt-realtime`

“Sure! The number is: 5-5-1-1-1-9-7-6-5-4-2-3. Please let me know if you need anything else!”

Instructions

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Identify the following issues:

- Ambiguity: Could any wording be interpreted in more than one way?
- Lacking Definitions: Are there any class labels, terms, or concepts that are not defined that might be misinterpreted by an LLM?
- Conflicting, missing, or vague instructions: Are directions incomplete or contradictory?
- Unstated assumptions: Does the prompt assume the model has to be able to do something that is not explicitly stated?

Do ****NOT**** list issues of the following types:

- Invent new instructions, tool calls, or external information. You do not know what tools need to be added that are missing.
- Issues that you are unsure about.

Output Format

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[END OF ISSUE]

Can you provide some variants of the prompt so that the model can better understand the constraints to alleviate the issue?

No Audio or Unclear Audio

Sometimes the model thinks it hears something and tries to respond. You can add a custom instruction telling the model on how to behave when it hears unclear audio or user input. Modify the desired behaviour to fit your use case (maybe you don't want the model to ask for a clarification, but to repeat the same question for example)

- **When to use:** Background noise, partial words, or silence trigger unwanted replies.
- **What it does:** Stops spurious responses and creates graceful clarification.

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Background Music or Sounds

Occasionally, the model may generate unintended background music, humming, rhythmic noises, or sound-like artifacts during speech generation. These artifacts can diminish clarity, distract users, or make the assistant feel less professional. The following instructions help prevent or significantly reduce these occurrences.

- **When to use:** Use when you observe unintended musical elements or sound effects in Realtime audio responses.
- **What it does:** Steers the model to avoid generating these unwanted audio artifacts.
- **How to adapt:** Adjust the instruction to try to explicitly suppress the specific sound patterns you are encountering.

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```
## lookup_account(email_or_phone)
...
```

```
## check_outage(address)
...
```

We need to ensure the tool list has the same availability tools and **the descriptions do not contradict each other:**

```
1  [
2  {
3      "name": "lookup_account",
4      "description": "Retrieve a customer account using
    either an email or phone number to enable verification and
    account-specific actions.",
```



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This is the responses after applying the instruction using `gpt-realtime`

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```
4     "description": "Retrieve a customer account using
5     either an email or phone number to enable verification and
6     account-specific actions.
7
8     Preamble sample phrases:
9     - For security, I'll pull up your account using the email
10    on file.
11    - Let me look up your account by {email} now.
12    - I'm fetching the account linked to {phone} to verify
13    access.
14    - One moment-I'm opening your account details."
15    "parameters": {
16        "...":
17        },
18        {
19            "name": "check_outage",
```

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– When calling a tool, do not ask for any user confirmation. Be proactive

This is the responses **after** applying the instruction using `gpt-realtime`

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Tool Call Performance

As use cases grow more complex and the number of available tools increases, it becomes critical to explicitly guide the model on when to use each tool and just as importantly, when not to. Clear usage rules not only improve tool call accuracy but also help the model choose the right tool at the right time.

- **When to use:** Model is struggling with tool call performance and needs the instructions to be explicit to reduce misuse.
- **What it does:** Add instructions on when to “use/avoid” each tool. You can also add instructions on sequences of tool calls (after Tool call A, you can call Tool call B or C)

Example

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Tip: If a tool call can fail unpredictably, add clear failure-handling instructions so the model responds gracefully.

Tool Level Behavior

You can fine-tune how the model behaves for specific tools instead of applying one global rule. For example, you may want READ tools to be called proactively, while WRITE tools require explicit confirmation.

- **When to use:** Global instructions for proactiveness, confirmation, or preambles don't suit every tool.
- **What it does:** Adds per-tool behavior rules that define whether the model should call the tool immediately, confirm first, or speak a preamble before the call.

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Confirmation phrase: "I can issue a credit for this outage—would you like me to go ahead?"

`## schedule_technician(account_id, window) – CONFIRMATION FIRST`

Use when: reboot + line checks fail AND outage=false.

Windows: "10am–12pm ET" or "2pm–4pm ET".

Confirmation phrase: "I can schedule a technician to visit—should I book that for you?"

`## escalate_to_human(account_id, reason) – PREAMBLES`

Use when: harassment, threats, self-harm, repeated failure, billing disputes > \$50, caller is frustrated, or caller requests escalation.

Preamble: "Let me connect you to a senior agent who can assist further."

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Examples

Example: raw string (more error-prone)

Tool returns:

I just sent you an email **with** the verification link. Please **open** it **and** click "Confirm". 

Model sometimes says:

- "I've emailed you a verification link..." (paraphrase)
- Drops the last sentence (truncation)
- Adds extra commentary ("Can I help with anything else?")

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awkward after receiving a thinker response.

- **What it does:** Adds clear instructions that guide the responder to rephrase the thinker's text into a short, natural, speech-first reply.
- **How to adapt:** Tweak phrasing style, openers, and brevity limits to match your use case expectation.

Example

```
# Tools
## Supervisor Tool
Name:
getNextResponseFromSupervisor(relevantContextFromLastUserMessage
: string)
```

When to call:



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up now.”).

- Keep it short: no more than 2 sentences.
- Use this template: opener + one-sentence gist + up to 3 key details + a quick confirmation or choice (for example: “Does that match what you expected?”, “Want me to review options?”).
- Read numbers for speech: money naturally (“\$45.20” → “forty-five dollars and twenty cents”), phone numbers 3-3-4, addresses with individual digits, dates/times plainly (“August twelfth”, “three-thirty p.m.”).

Here’s an example without the rephrasing instruction:

“Assistant: Your current credit card balance is positive at 32,323,232 AUD.”

Here’s the same example with the rephrasing instruction:

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the session or doesn't want to continue. If it's ambiguous, confirm with the customer before calling.

Conversation Flow

This section covers how to structure the dialogue into clear, goal-driven phases so the model knows exactly what to do at each step. It defines the purpose of each phase, the instructions for moving through it, and the concrete “exit criteria” for transitioning to the next. This prevents the model from stalling, skipping steps, or jumping ahead, and ensures the conversation stays organized from greeting to resolution.

As well, by organizing your prompt into various conversation states, it becomes easier to identify error modes and iterate more effectively.

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(for billing) are known.

3) Verify

Goal: Confirm identity and retrieve the account.

How to respond:

- Once you have email or phone, call `lookup_account(email_or_phone)`.
- If lookup fails, try the alternate identifier once; otherwise proceed with general guidance or offer escalation if account actions are required.

Exit when: Account ID is returned.

4) Diagnose

Goal: Decide outage vs local issue.

How to respond:

- For connectivity, call `check_outage(address)`.

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Sample phrases act as “anchor examples” for the model. They show the style, brevity, and tone you want it to follow, without locking it into one rigid response.

- **When to use:** Responses lack your brand style or are not consistent.
- **What it does:** Provides sample phrases the model can vary to stay natural and brief.
- **How to adapt:** Swap examples for brand-fit; keep the “do not always use” warning.

Example

Sample Phrases

– Below are sample examples that you should use for inspiration.
DO NOT ALWAYS USE THESE EXAMPLES, VARY YOUR RESPONSES.



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SERVICE:

– “Hi there—tell me what you’d like help with.”

Exit when: Caller states an initial goal or symptom.

2) Discover

Goal: Classify the issue and capture minimal details.

How to respond:

- Determine billing vs connectivity with one targeted question.
- For connectivity: collect the service address.
- For billing/account: collect email or phone used on the account.

Sample phrases (do not always repeat the same phrases, vary your responses):

- “Is this about your bill or your internet speed?”
- “What address are you using for the connection?”
- “What’s the email or phone number on the account?”

Exit when: Intent and address (for connectivity) or email/phone (for billing) are known.

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Exit when: Root cause known.

5) Resolve

Goal: Apply fix, credit, or appointment.

How to respond:

- If confirmed outage > 240 minutes in the last 7 days, call `refund_credit(account_id, 60)`.
- If outage=false and issue persists after basic checks, offer "10am-12pm ET" or "2pm-4pm ET" and call `schedule_technician(account_id, chosen window)`.
- If the local fix worked, state the result and next steps briefly.

Sample phrases (do not always repeat the same phrases, vary your responses):

- "There's been an extended outage-adding a 60-minute bill credit now."
- "No outage-let's book a technician. I can do 10am-12pm ET or

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stay focused and responsive, which improves user experience.

Two common patterns for managing complex scenarios are:

1. Conversation Flow as State Machine
2. Dynamic Conversation Flow via session.updates

Conversation Flow as State Machine

Define your conversation as a JSON structure that encodes both states and transitions. This makes it easy to reason about coverage, identify edge cases, and track changes over time. Since it's stored as code, you can version, diff, and extend it as your flow evolves. A state machine also gives you fine-grained control over exactly how and when the conversation moves from one state to another.

Example

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```
25     "Politely ask, 'who do I have the pleasure of
speaking with?'",
26     "Do NOT verify or spell back the name; just accept
it."
27 ],
28     "examples": [
29         "Who do I have the pleasure of speaking with?"
30     ],
31     "transitions": [{
32         "next_step": "3_get_and_verify_phone",
33         "condition": "Once name is obtained, OR name is
already provided."
34     }]
35 },
36 {
37     "id": "3_get_and_verify_phone",
38     "description": "Request phone number and verify by
repeating it back.",
```

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This approach reduces the model's cognitive load, making it easier for it to handle complex tasks without being distracted by unnecessary context.

Example

```
1 from typing import Dict, List, Literal
2
3 State = Literal["verify", "resolve"]
4
5 # Allowed transitions
6 TRANSITIONS: Dict[State, List[State]] = {
7     "verify": ["resolve"],
8     "resolve": [] # terminal
9 }
10
```



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```
    "string"}},
40         "required": ["email_or_phone"]
41     }
42 },
43 "resolve": [{
44     "type": "function",
45     "name": "schedule_technician",
46     "description": "Book a technician visit.",
47     "parameters": {
48         "type": "object",
49         "properties": {
50             "account_id": {"type": "string"},
51             "window": {"type": "string", "enum": ["10-
12 ET", "14-16 ET"]}
52         },
53         "required": ["account_id", "window"]
54     }
55 }]
```

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```
85         "instructions": INSTRUCTIONS_BY_STATE[state],  
86         "tools": TOOLS_BY_STATE[state] +  
            [build_state_change_tool(state)]  
87     }  
88 }
```

Safety & Escalation

Often with Realtime voice agents, having a reliable way to escalate to a human is important. In this section, you should modify the instructions on WHEN to escalate depending on your use case.

- **When to use:** Model is struggling in determining when to properly escalate to a human or fallback system
- **What it does:** Defines fast, reliable escalation and what to say.

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USER

Hey, I don't want to talk to a bot, transfer me.

ASSISTANT

I understand you'd like to speak to a human. Could you please let me know what the issue is, so I can transfer you to the right specialist?

This is the conversation responses from our new GA model `gpt-realtime` using the instruction.

USER

I don't want to talk to a bot, can you transfer me?

ASSISTANT 🗣️

I'm sorry for the trouble—I'm transferring you to a specialist now.

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